

Exam 1 review day

Reminders and agenda

- + Exam 1 on Friday
 - + Bring a calculator!
 - + Review questions and solutions on the course website
- + Today:
 - + Quick discussion about outliers activity from Monday
 - + Exam 1 review

Outliers

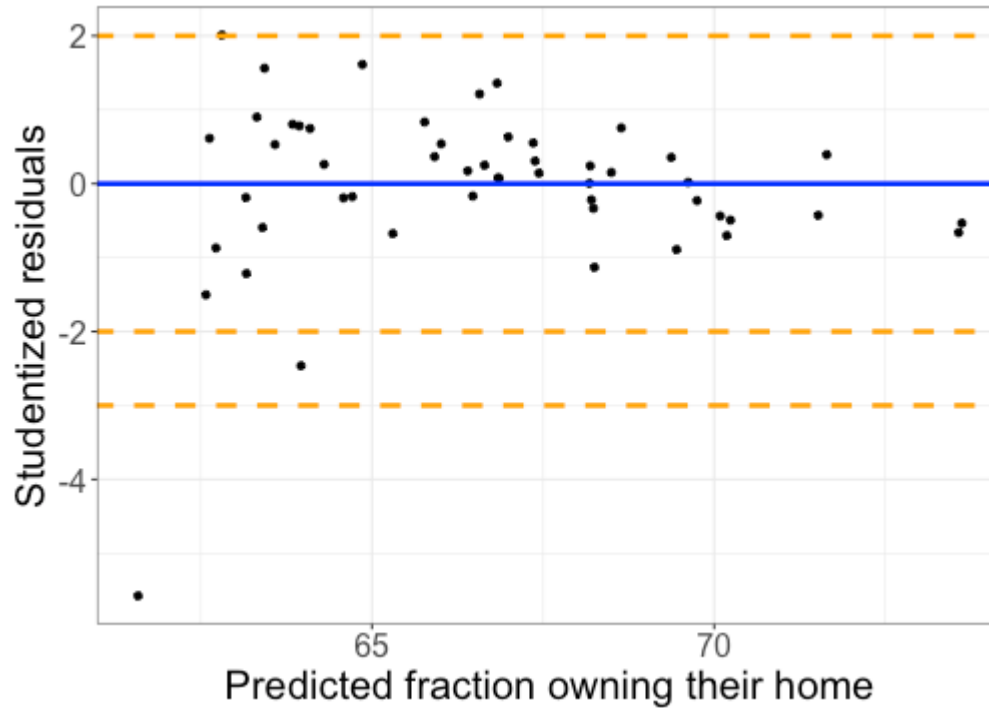
When can we *remove* potential outliers from the data?

- + When the observation is a clear error (e.g., a negative value for height)
- + When the observation comes from a different population than our population of interest (e.g., if we want to fit a model about dogs, then we can remove a cat from the data)

In all other situations:

- + Report the results (estimated coefficients, confidence intervals, test statistics, etc.) both **with** AND **without** the potential outlier(s)

Last time: exploring outliers



What point is the outlier here?

Last time: exploring outliers

DC: Percent urban -- 100%, Percent owning home -- 42%

Is this observation a clear *error*?

Last time: exploring outliers

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No -- these are believable numbers, given what we know about DC

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Does DC belong to a different population than our population of interest?

Last time: exploring outliers

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Is this observation a clear *error*?

No -- these are believable numbers, given what we know about DC

Does DC belong to a different population than our population of interest?

Maybe -- I think you can make an argument either way

Last time: exploring outliers

How should we handle DC?

- + If we think DC does not belong in our population of interest, could remove
- + Otherwise, report results with and without DC